

The encoder, not the bins: decomposing peak-level MS/MS gains

Xulin Pan^{1*} Chuanming Wang²

¹The Pennsylvania State University, University Park, PA, USA

²School of Biological and Food Engineering, Honghe University, Mengzi, China

Abstract

Machine-learning methods for tandem mass spectrometry (MS/MS) have moved away from fixed-width binned intensity vectors towards models that read peaks at full precision, and the resulting gains are commonly attributed to the spectral representation. That attribution has never been tested, because abandoning bins forces an architecture change at the same time: a peak set cannot be read by the network that reads a binned vector. Here we separate them with a rounding control that snaps each peak to its bin centre, merges the collisions, and passes the result through the *identical* peak-set Transformer at an identical parameter count, so each of the encoder and the mass axis — the precision of the m/z scale — changes with the other held constant. On an openly licensed corpus of 414,992 spectra from GNPS, MassBank and MoNA spanning 66 instrument types, over 10,058 held-out molecules and three seeds, replacing a 0.5 Da binned baseline with a full-precision peak-set model raised macro AUPRC from 0.1130 to 0.1709 (chance 0.0085). The control splits that paired gain of +0.0579 into +0.0438 (95% CI +0.0364 to +0.0517) for the encoder and +0.0142 (+0.0097 to +0.0188) for the mass axis: 76% is architectural, and the ranking encoder > aggregation > mass axis holds in every seed. Repeating the ladder on a single-library corpus that is 99% one instrument gives the encoder 58%. Improvements measured against binned baselines should not be read as evidence about spectral representation. Code: <https://github.com/xulinpan/cheminformatics>.

Scientific contribution

We introduce a rounding control that isolates the spectral representation from the encoder in MS/MS substructure prediction. Existing benchmarks compare featurizations bundled with the encoders that read them, because changing the representation ordinarily forces an architecture change; the control removes that coupling. Across three seeds and two corpora of sharply different acquisition diversity, the encoder accounts for 76% and 58% of the gain a two-arm comparison credits to the representation, and is in both cases the largest of the three effects we measure. The consequence for practice is that performance measured against a binned baseline confounds representation with architecture and cannot support claims about either alone.

Keywords

Tandem mass spectrometry, Structure annotation, Spectral representation, Mass defect, Substructure prediction, Ablation study

*Correspondence: xulinpanias@gmail.com

Introduction

Assigning a chemical structure to an MS/MS spectrum is the rate-limiting step in untargeted metabolomics. Where the molecule appears in a reference library the problem reduces to spectral matching, but most spectra acquired from biological extracts do not [1]. For the remainder, computational annotation relates the spectrum to candidate structures: by in-silico fragmentation and scoring, as in MetFrag [2]; by predicting a molecular fingerprint and searching a structure database, as in CSI:FingerID [3] and MIST [4]; or by learning a spectrum representation whose geometry reflects chemical similarity, as in MS2DeepScore [5] and DreaMS [6].

For most of the past decade these systems took the same input object. The spectrum, physically a sparse list of $(m/z, \text{intensity})$ pairs, was discretised onto a fixed grid of bins and handed to the model as a dense vector. The convention came from spectral library search, where a bin comfortably wider than the instrument’s mass error makes cosine matching robust to calibration drift between laboratories. A 0.5 Da bin spans roughly 1,700 ppm at m/z 300 while a modern time-of-flight instrument delivers about 5 ppm, so everything the instrument resolves inside that window is discarded — including the mass defect, the difference between a fragment’s exact and nominal mass, which distinguishes C_5H_7^+ (67.0542) from $\text{C}_4\text{H}_3\text{O}^+$ (67.0178) and therefore carries elemental composition.

The field has since moved away from naive binning. CSI:FingerID scores against fragmentation trees over exact masses, MIST tokenises peaks by chemical formula, DreaMS operates on peak lists, and TransExION aligns fragments by nominal mass with the defect embedded as a separate channel [14]; the direction is surveyed in [11], and the convention of discretising a spectrum onto a fixed grid persists well beyond MS/MS [13], and how best to present a spectrum to a model remains open enough that serialising peaks as text for a language model is an active line of enquiry [12]. These systems outperform binned baselines, and the improvement is routinely read as evidence that the representation was the problem.

Binning has not gone unexamined. Giné et al. [7] benchmark featurization strategies for structure annotation across a large grid of models, comparing adaptive binning and spectrum hashing against learned embeddings, and report that retrieval degrades markedly as the mass tolerance is relaxed. Working on chemical-similarity prediction rather than structure annotation, de Jonge et al. [8] settle on 0.1 Da bins and report that finer bins did not improve performance. These are the two most direct empirical statements available about what mass precision is worth, and they do not agree with each other — which is itself a reason to ask the question in a form that admits a clean answer.

That reading does not follow from the experiments, and the reason is structural rather than careless. A binned vector and a peak set are different kinds of object: the first is dense and fixed-length, the second sparse and variable-length and permutation-invariant. No architecture reads both. Abandoning bins therefore forces a change of encoder, and a comparison of a peak-level model against a binned baseline varies the representation and the architecture together, usually alongside differences in training data and objective.

This applies to benchmarks as well as to individual systems. A study that compares many featurizations [7] measures each one bundled with the encoder that reads it — adaptive binning with a network that takes a dense vector, a learned embedding with the architecture that produced it — and so estimates the joint effect of representation and architecture at many points rather than separating them at any one. That is the right design for choosing a pipeline and the wrong one for

attributing a gain. What is missing is not breadth but a contrast in which the mass axis moves and nothing else does.

This paper supplies the separation. The obstacle above is real but not insurmountable: while a peak-set encoder cannot read a binned vector, it can read peaks whose m/z has been *rounded to the bin grid*. Quantising the mass axis — we use the term throughout for the precision of the m/z scale, the component of the input representation that binning actually destroys — while leaving the peak-set interface intact yields an arm that carries the baseline’s mass information through the treatment’s architecture, and inserting it between the two turns a confounded two-arm comparison into a decomposition. We report that decomposition across three training seeds, extend it with a bin-width sweep that prices precision as a continuous quantity, and examine what the results imply for how baselines in this area should be constructed and read.

Methods

Corpora and splits

Both corpora are drawn from the CASMI 2026 training corpus, an aggregation of eleven public and in-house MS/MS libraries whose `ingest_lib` column records each spectrum’s source.

Our primary corpus is the openly licensed subset: every spectrum contributed by GNPS, MassBank or MoNA, 414,992 spectra over 52,428 structures, spanning 66 instrument types and 112 adducts. We use it because it is the part of the corpus a third party can obtain from its sources without an account, and because it is acquisitionally diverse.

Our secondary corpus is a contiguous slice of two of the file’s twenty-one row groups, 262,144 spectra. Because the file is ordered by source library, that slice is not a sample of the corpus: it is 99.0% one in-house library and 1.0% another, and 99% of its spectra come from a single instrument (timsTOF), across two instrument types and sixteen adducts. We report it not as a second sample but as a deliberately contrasting one — a near-homogeneous corpus against a heterogeneous one — which is what makes the comparison in Fig. 2 informative.

Spectra were retained if the precursor m/z was below 2,000 and the reported precursor error within 20 ppm; peaks were retained above 0.1% of the base peak and below the precursor plus 1.5 Da, capped at the 256 most intense, with m/z held in double precision throughout preprocessing. On the open corpus this leaves 372,200 spectra over 50,287 structures and 24,627,256 peaks.

Structures were assigned to five folds by grouping generic Mureko scaffolds, with acyclic structures falling back to grouping by their own identifier, and fold sizes balanced greedily. Scaffold grouping is stricter than a random split and follows the generalisation-demanding practice argued for in MassSpecGym [9]; we did not adopt that benchmark’s own split because our corpora are selected by source library rather than drawn from it. Folds 2–4 were used for fitting, fold 1 for model selection, and fold 0 was held out; all reported numbers are on fold 0 (10,058 molecules on the open corpus, 8,692 on the single-library corpus).

Structural targets

Each structure was decomposed with BRICS and represented by the presence or absence of each of the 512 most frequent attachment-labelled fragments, the vocabulary fitted on the training folds alone. On the open corpus the target matrix is 99.3% zero with a median of 4 positive targets per molecule; 433 targets carry at least five held-out positives, at a mean prevalence of 0.0085, which is the chance level for macro AUPRC. The single-library corpus is less sparse (97.1% zero, median 17,

477 scored targets, chance 0.0093), reflecting how much more chemical variety three public libraries carry than one in-house library. Because almost all targets fall below 1% prevalence we report AUPRC as the primary metric and AUROC only as a secondary summary.

The four arms

The design is a ladder in which each arm differs from the one below it in exactly one respect.

M0, the conventional baseline. Each spectrum is projected onto 2,048 fragment bins and 2,048 precursor-minus-fragment bins of width 0.5 Da over $[0, 1024)$, with square-root intensity accumulated into each bin and the concatenated vector L_2 -normalised. The 4,096-dimensional vector is encoded by a two-layer perceptron. This is the representation used, with minor variation in bin width, throughout the MS/MS machine-learning literature.

M1R, the rounding control. Each peak’s m/z is assigned to a cell by $\lfloor m/w \rfloor$ with $w = 0.5$ Da — the same indexing M0’s encoder uses — and placed at that cell’s centre. Peaks sharing a cell are merged by accumulating square-root intensity, the accumulator M0 applies, so the merged peak carries the total M0 would have seen. The resulting tokens pass through the peak-set encoder described below. The arm therefore holds mass information at M0’s resolution while using M1’s architecture.

One detail is load-bearing. The encoder derives the neutral loss as $m^{\text{prec}} - m$. Quantising the peaks while leaving the precursor at full precision would return the precursor’s mass defect to every token through that subtraction, silently defeating the control, so the precursor is snapped to the same grid whenever quantisation is active. A unit test asserts that two precursors in one cell give identical embeddings under the control and different embeddings without it.

M1, full precision. Identical to M1R with quantisation disabled, so peaks reach the encoder at the precision the instrument reported. M1R and M1 are identical in every other configuration field and have identical parameter counts (712,336); a test asserts both.

M2, hierarchical aggregation. Identical to M1 except that mean pooling over a molecule’s spectra is replaced by a measurement model weighting each spectrum by a learned, acquisition-dependent precision. This arm is included to give the other two effects a scale, not as a contribution of this paper.

Peak encoding

Each peak is encoded as a token. For a peak at m/z m in a spectrum with precursor m^{prec} we form a multi-scale sinusoidal embedding

$$\varphi_L(m) = [\sin(2\pi m/\lambda_1), \cos(2\pi m/\lambda_1), \dots, \sin(2\pi m/\lambda_L), \cos(2\pi m/\lambda_L)]^\top, \quad (1)$$

with $L = 32$ wavelengths spaced geometrically from 2 mDa to 10^3 Da, applied both to m and to the neutral loss. Short wavelengths resolve the mass defect and long wavelengths carry nominal mass. The token also receives the explicit defect $\delta = m - \text{round}(m)$, square-root and log-transformed intensity, and intensity rank.

At a wavelength of 2 mDa and m/z 342 the phase in Eq. (1) exceeds 10^6 , and evaluating it in single precision destroys exactly the low-order information the embedding exists to preserve. We therefore compute the phase in double precision and reduce it modulo 2π before casting. With this reduction the embeddings of $\text{C}_4\text{H}_3\text{O}^+$ and C_5H_7^+ , which share a 0.5 Da bin, differ by 1.82 in the infinity norm.

Tokens are encoded by a four-block Transformer with a learned pooling token, each block

conditioned on the acquisition covariates (adduct, polarity, instrument family, harmonised collision energy) by adaptive layer normalisation. The encoder is permutation invariant and accepts variable peak counts.

Bin-width sweep

To price precision as a continuous quantity rather than at one grid, we repeated the M1R construction at $w \in \{0.25, 0.1, 0.05, 0.01\}$ Da, with the encoder, parameter count, schedule and seed fixed. Together with $w = 0.5$ (M1R) and full precision (M1) this gives six points differing only in how finely the mass axis is resolved. The sweep was run on the single-library corpus only.

Training protocol

Every arm was trained for fifteen epochs with AdamW, a one-cycle schedule peaking at 10^{-3} , batches of 16 molecules, mixed precision and gradient clipping at 1.0. The loss was binary cross-entropy over the 512 presence targets with an auxiliary negative-binomial term on 88 functional-group counts, and checkpoints were selected on validation macro AUPRC. No hyperparameter search was performed for any arm. The four ladder arms were each trained three times under seeds $\{2026, 7, 13\}$ on each corpus — twenty-four training runs in all — and the sweep arms once each.

Evaluation

We report macro AUPRC over targets with at least five held-out positives, with macro AUROC, micro AUPRC and Brier score as secondary summaries. Because arms are evaluated on identical molecules and identical targets, the appropriate contrast is paired: each target’s average precision is first averaged over the three seeds, the per-target differences are then taken, and their mean is bootstrapped over targets (5,000 resamples). We report the fraction of targets improved alongside, and, for each effect, its standard deviation across seeds — which, unlike the bootstrap interval, reflects initialisation variance. Where an ordering between effects is at issue we check it within each seed rather than reading it off the means.

Results

The four arms

The design is a ladder in which each arm differs from the one below it in exactly one respect (Table 1). M0 is the conventional baseline: a 0.5 Da binned intensity vector read by a two-layer perceptron. M1R keeps that binned mass axis but replaces the perceptron with the peak-set Transformer, snapping each peak to its bin centre and merging the peaks that collide, so the encoder receives tokens carrying no more mass information than the binned vector did. M1 hands the same Transformer the peaks at full precision. M2 adds hierarchical pooling over a molecule’s spectra. M1R and M1 are identical in every configuration field but the width of the mass grid, and carry identical parameter counts. Each arm was trained three times, under seeds $\{2026, 7, 13\}$.

M1R is the rung that makes the experiment mean anything. Without it, the contrast between a binned baseline and a peak-set model changes the mass axis and the encoder together, and a difference between them cannot be assigned to either.

Three quarters of the apparent representation gain is the encoder

Replacing the binned vector with a full-precision peak set raises macro AUPRC from 0.1130 to 0.1709 against a chance level of 0.0085, a paired per-target gain of +0.0579 (95% CI +0.0494 to

Table 1: The four arms on the open corpus, each differing from the one above it in a single respect. Macro AUPRC on 10,058 held-out molecules, mean over three seeds. Chance is 0.0085 over 433 scored targets.

arm	mass axis	encoder	pooling	parameters	macro AUPRC	SD over seeds
M0	0.5 Da bins	MLP	mean	1,164,560	0.1130	0.0012
M1R	0.5 Da bins	Transformer	mean	712,336	0.1567	0.0025
M1	full precision	Transformer	mean	712,336	0.1709	0.0022
M2	full precision	Transformer	hierarchical	829,584	0.1888	0.0015

+0.0667) improving 81.8% of targets — the effect a two-arm comparison would report. The control splits it (Fig. 1, Table 2). Changing the encoder alone, with the mass axis binned on both sides, is worth +0.0438 (+0.0364 to +0.0517; 79.2% of targets). Changing the mass axis alone, with the same encoder on both sides, is worth +0.0142 (+0.0097 to +0.0188; 64.9% of targets). The two components sum to the joint effect exactly, as they must, since each target’s contribution enters both sides linearly.

Seventy-six per cent of what a two-arm comparison would credit to the spectral representation is therefore the encoder. Both components are real — each interval excludes zero and each improves a clear majority of targets — but the larger by a factor of three is the one that has nothing to do with how the spectrum is represented.

Table 2: Paired per-target differences on the open corpus, bootstrapped over targets after averaging each target’s average precision across the three seeds. The final column is the standard deviation of the effect across seeds.

contrast	isolates	Δ AUPRC	95% CI	targets improved	SD over seeds
M1R – M0	encoder	+0.0438	[+0.0364, +0.0517]	79.2%	0.0030
M1 – M1R	mass axis	+0.0142	[+0.0097, +0.0188]	64.9%	0.0004
M2 – M1	aggregation	+0.0179	[+0.0137, +0.0222]	63.7%	0.0009
M1 – M0	encoder and mass axis	+0.0579	[+0.0494, +0.0667]	81.8%	–
M2 – M0	all three	+0.0758	[+0.0656, +0.0862]	83.4%	–

The ordering is stable across seeds

All three pairwise orderings hold in every seed. The encoder effect exceeds the mass-axis effect by +0.0331, +0.0288 and +0.0268, and the aggregation effect by +0.0303, +0.0243 and +0.0231; aggregation exceeds the mass axis by +0.0028, +0.0045 and +0.0038. The ranking encoder > aggregation > mass axis therefore holds three times out of three, and the mass-axis effect is the most stable of the three across seeds (SD 0.0004, against 0.0030 for the encoder).

That the smallest effect is also the most reproducible is worth stating plainly: the mass axis is not a marginal effect lost in noise, it is a small effect measured precisely.

The same decomposition on a very different corpus

Because the finding is a claim about how the field should construct comparisons rather than about one dataset, we repeated the entire ladder on a second corpus with almost nothing in common with the first: a single in-house library, 99% of whose spectra come from one instrument (timsTOF), against the open corpus’s 66 instrument types drawn from three public libraries (Fig. 2, Table 3).

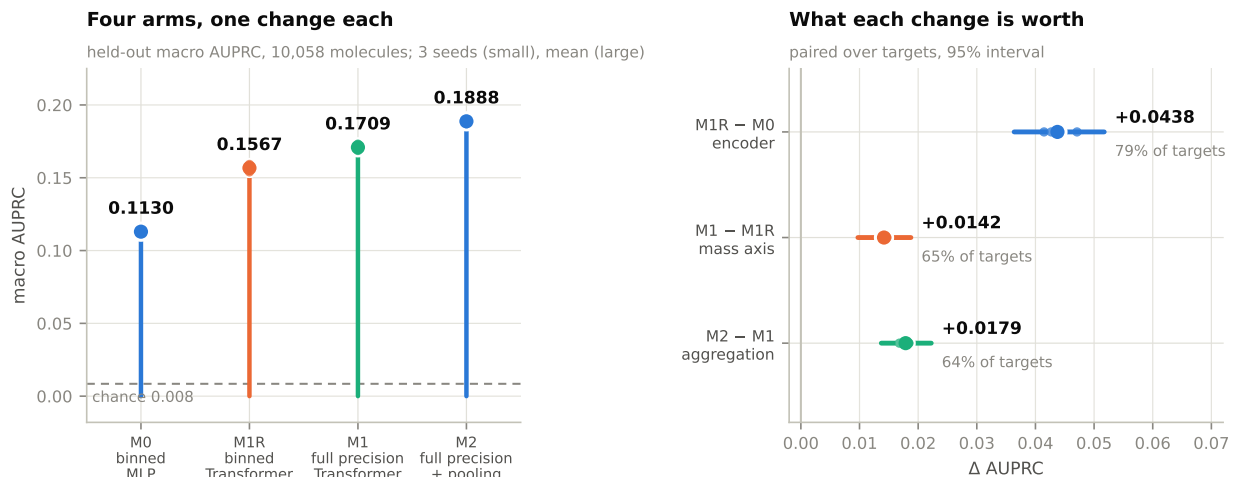


Figure 1: Decomposing the gain on the open corpus. **Left:** held-out macro AUPRC for the four arms; small dots are individual seeds, large dots their mean. **Right:** what each single change is worth, paired over targets with 95% bootstrap intervals; small dots are the per-seed effect sizes.

The qualitative result is unchanged — the encoder dominates on both — but the balance shifts. On the single-library corpus the encoder accounts for 58.5% of the joint gain; on the open corpus, 75.5%. The mass-axis effect falls from +0.0268 to +0.0142, roughly halving, while the encoder effect grows slightly from +0.0377 to +0.0438.

The single-library corpus also leaves one ordering unresolved that the open corpus settles. There, the mass axis averaged +0.0268 against aggregation’s +0.0322, but the sign of that difference reversed under one of three seeds, and we could not rank them. On the open corpus the same comparison holds in all three seeds.

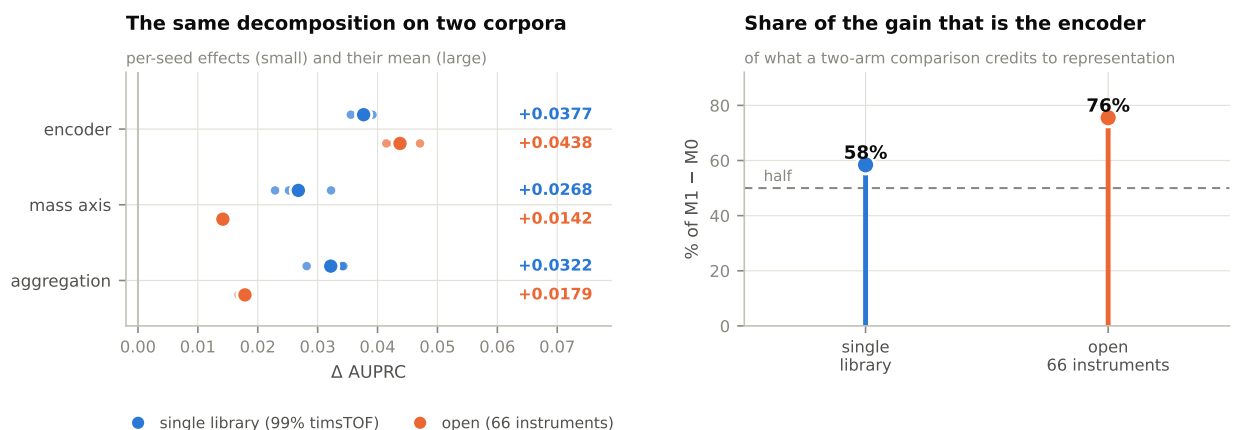


Figure 2: The decomposition on two corpora. **Left:** each effect measured on both, with per-seed values (small) and their mean (large). **Right:** the encoder’s share of the gain a two-arm comparison would attribute to the representation.

What binning removes, and how much of it matters

The mechanism is visible before any model is fitted. Of the 3,797,166 peaks in the open corpus’s

Table 3: The decomposition on two corpora. The single-library corpus is 99% timsTOF from one in-house library; the open corpus spans 66 instrument types across GNPS, MassBank and MoNA. Absorption is measured on each corpus’s own held-out spectra.

effect	single library	SD (single)	open corpus	SD (open)
encoder	+0.0377	0.0019	+0.0438	0.0030
mass axis	+0.0268	0.0049	+0.0142	0.0004
aggregation	+0.0322	0.0035	+0.0179	0.0009
encoder share of M1 – M0	58.5%	–	75.5%	–
peaks absorbed at 0.5 Da	34.4%	–	23.6%	–

held-out spectra, 23.6% fall into a 0.5 Da bin already occupied by another peak from the same spectrum and are therefore absorbed. On the single-library corpus’s held-out spectra the same figure is 34.4% (Fig. 3). The loss falls with the grid on both, reaching 2.9% and 6.5% respectively at 0.01 Da.

What a merged peak costs is compositional identity. Enumerating plausible CHNOS compositions within 10 ppm places 328 candidate fragment formulas inside a single 0.5 Da bin at m/z 300; the count is governed by the width of the mass window in the way Kind and Fiehn established for precursor formulas [10], so a bin three orders of magnitude wider than the instrument’s error discards essentially all of the discrimination the measurement bought.

The two corpora therefore differ in the same direction on both quantities: the one that loses more peaks to binning is also the one where the mass axis is worth more (34.4% and +0.0268; 23.6% and +0.0142). The correspondence is not proportional — absorption falls to 0.69 of its value while the effect falls to 0.53 — so peak absorption is not a complete account of the cost, but it moves with it.

We offer one hypothesis for the difference, which our design does not test. The single-library corpus is almost entirely timsTOF, a high-resolution instrument on which every reported m/z carries real precision. The open corpus spans 66 instrument types of varying resolution, and on the lower-resolution ones there is less precision present to preserve, so the average value of preserving it falls. Stratifying the mass-axis effect by instrument resolution would test this directly and we regard it as the most informative experiment this work leaves undone.

How much precision is needed

The rounding control prices one particular grid. Sweeping the grid width on the single-library corpus, with the encoder and parameter count fixed throughout, prices precision as a function of how much of it is retained (Table 4). At 0.5, 0.25 and 0.1 Da the model sits between 0.0801 and 0.0867, a range of 0.0066 — under twice that arm’s seed standard deviation of 0.0039, and unordered within itself; coarsening from 0.1 to 0.5 Da costs nothing measurable. Below that the curve moves: 0.05 Da reaches 0.0909 and 0.01 Da reaches 0.1030, closing 65% of the distance to full precision. Even a 10 mDa grid — fine enough to separate the isobaric fragments that motivate the mass-defect argument — leaves roughly a third of the benefit unrealised, which suggests the model uses the defect as a continuous quantity rather than merely as a discriminator between candidate formulas.

Two cautions. Each width was run once, so the non-monotonic dip at 0.25 and 0.1 Da, spanning 0.0066, is unremarkable across five single runs and we read it as noise rather than structure. We

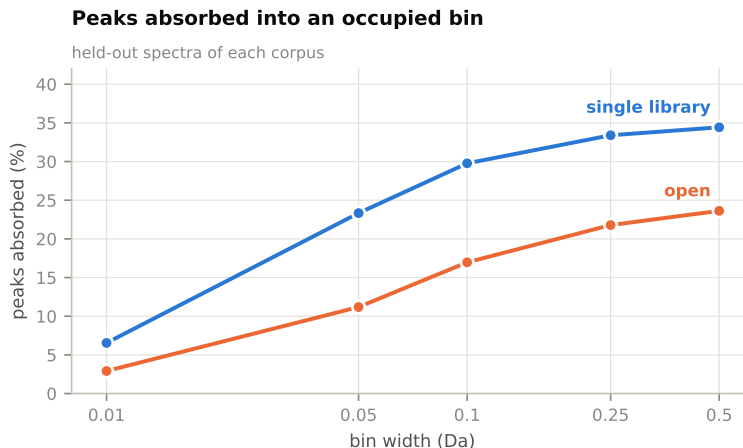


Figure 3: Fraction of peaks absorbed into an already-occupied bin, against grid width, on each corpus’s own held-out spectra.

tested whether it might instead be an aliasing artefact — the sinusoidal basis has wavelengths that are integer divisors of these grid spacings, which would render some channels constant — but only one to three of thirty-two channels are affected at any width and the count does not track performance ($r = -0.26$), so we discount that explanation. The sweep was run on the single-library corpus only; whether the 0.05 Da threshold moves with instrument diversity is open.

Table 4: Bin-width sweep, single seed. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. The final column expresses each result as a share of the distance between the 0.5 Da arm and full precision.

bin width (Da)	peaks absorbed	macro AUPRC	share of the 0.5 Da gap closed
0.5	37.5%	0.0867	0%
0.25	36.8%	0.0818	-19%
0.1	33.1%	0.0801	-26%
0.05	25.9%	0.0909	17%
0.01	7.5%	0.1030	65%
full precision	0.0%	0.1119	100%

Discussion

The headline is a reattribution. A two-arm comparison on the open corpus credits the spectral representation with +0.0579 macro AUPRC; once the encoder is held constant, the representation’s share is +0.0142 and the encoder’s is +0.0438. Both are real, and the smaller by a factor of three is the one the literature has been discussing.

It is worth being clear about why this went unnoticed, because the answer is not inattention. The confound is forced by the data structures involved. A binned vector is dense and fixed-length; a peak set is sparse, variable-length and permutation-invariant. No architecture reads both, so an experiment that changes only the representation appears impossible, and the comparison everyone can run is the one that changes both. The rounding control works not by finding a network that

reads both objects but by degrading one object into the other while keeping the interface fixed. The manoeuvre is simple in retrospect, and we suspect it generalises: wherever a representation change forces an architecture change, asking whether the richer representation can be impoverished in place is likely to be more productive than looking for a universal encoder.

Our result does not diminish the systems that abandoned binning. CSI:FingerID, MIST, DreaMS and TransExION are better than binned baselines, and this work does not dispute it. What it changes is the account of *why*. To the extent that those systems differ from a binned baseline in both respects, our decomposition suggests the larger part of their advantage comes from the architectures they adopted rather than from the mass information they preserved. That is a testable claim about each of them, and the control we describe is cheap enough to apply.

The two corpora sharpen this and add a caveat of their own. The encoder dominates on both, so the conclusion does not depend on a single dataset — but its margin does. On a corpus that is 99% one high-resolution instrument the encoder takes 58% of the gain; on one spanning 66 instrument types it takes 76%, and the mass axis roughly halves. Our hypothesis is that a heterogeneous corpus contains many spectra whose reported m/z carries little real precision, so preserving precision helps less on average. If that is right, the value of full-precision input depends on the instrument mix a method will encounter, and a method tuned on a homogeneous high-resolution corpus will disappoint on a heterogeneous one for reasons that have nothing to do with its architecture. We did not test this, and say so in the Limitations; we flag it because it bears directly on how these methods should be benchmarked.

The sweep adds a second complication. Precision buys nothing measurable between 0.5 and 0.1 Da; the gain appears below 0.05 Da and is not exhausted at 0.01 Da. A 10 mDa grid separates the isobaric fragments that motivate the mass-defect argument, yet still leaves about a third of the benefit unrealised, which suggests the model uses the defect as a continuous quantity rather than merely as a discriminator between candidate formulas. This sits awkwardly with the common design choice, TransExION’s included, of embedding the defect alongside a coarsely quantised nominal mass: on our evidence the resolution of that channel matters more than its presence.

It also disagrees with a recent result we should not paper over. de Jonge et al. [8] adopt 0.1 Da bins for MS2DeepScore 2.0 and report that finer bins did not improve performance; our sweep finds nothing measurable between 0.5 and 0.1 Da, agreeing with them over that range, but a gain of +0.0121 between 0.05 and 0.01 Da, disagreeing below it. Three explanations seem live. The tasks differ — chemical-similarity prediction between two spectra may be satisfied by coarser features than inferring which substructures a molecule contains, since the former needs only that two spectra of similar molecules land near each other. The architectures differ. And our sweep is single-seed, so the disagreement may be ours. We cannot adjudicate this from our data, and flag it because a reader comparing the two papers will notice, and because the question of what bin width suffices is evidently not settled by either.

The practical consequence concerns controls. A method evaluated against a binned baseline is compared against an arm handicapped in two ways at once, and on this evidence the larger handicap is architectural. A paper reporting a gain over such a baseline has not shown that its representation is responsible, whatever its discussion says. The remedy is inexpensive: report the comparator’s encoder as explicitly as its representation, and where the claim is about representation, include an arm that changes only that.

Limitations

The open corpus is roughly a sixth of the available spectra and the single-library corpus a tenth; absolute values will move with scale even if the ordering does not. Three seeds per arm establish the ordering on the open corpus, where it holds in every seed, but on the single-library corpus the mass axis and aggregation could not be separated, and we declined to rank them there. The sweep carries one run per width and was run on one corpus, so only its endpoints should be read as findings and its 0.05 Da threshold may not transfer. The instrument-diversity hypothesis above is untested: stratifying the mass-axis effect by instrument resolution would settle it and is the most informative experiment this work leaves undone. Finally, and most substantively, M1R does not separate dense from sparse encoding: a Transformer over bin-indexed tokens is not a perceptron over a dense vector, and some of what we attribute to the encoder may be that switch rather than the attention mechanism. A fifth arm — a perceptron over the rounded peak set, or a Transformer over bin tokens without merging — would resolve it, and we regard that as the natural next experiment rather than a caveat that undermines the present one.

Conclusion

Binned intensity vectors entered MS/MS machine learning from spectral library search, and the field’s move away from them has been read as evidence that the spectral representation was the limiting factor. That reading was never tested, because no architecture reads both a binned vector and a peak set, so representation and encoder have always changed together. Inserting an arm that carries binned mass information through a peak-set encoder separates them. On an openly licensed corpus of 414,992 spectra spanning 66 instrument types, across three seeds and 10,058 held-out molecules, 76% of the gain attributed to the representation is the encoder (+0.0438) and 24% is the mass axis (+0.0142), with the ranking encoder > aggregation > mass axis holding in every seed. On a near-homogeneous single-library corpus the same decomposition gives 58% and 42%. Mass precision matters, and matters more the more homogeneous and higher-resolution the instruments are, but on both corpora it matters less than the architecture adopted alongside it. Improvements measured against binned baselines cannot be used as evidence about either.

Supplementary Information

Additional file 1 reports per-target average precision for every arm and seed, the full bootstrap distributions, the per-seed effect sizes, and the peak-absorption analysis broken down by adduct and collision energy.

Acknowledgements

We thank the CASMI 2026 organisers for assembling and releasing the corpus.

Author contributions

X.P. designed the study, implemented the models and the control, performed the experiments and analyses, and wrote the manuscript. C.W. contributed to the interpretation of the chemical results

and revised the manuscript. Both authors read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

All code, configurations, trained-run reports and the analysis that generates every number in this paper are available at <https://github.com/xulinpan/cheminformatics> under the MIT licence, including the script that builds the open subset and a document recording each source library’s contribution. The primary corpus consists of the spectra contributed by GNPS, MassBank and MoNA, all of which are openly licensed and obtainable from those sources; the aggregated CASMI 2026 file from which we selected them is distributed by the competition organisers, and the in-house libraries it also contains are not redistributable and are not used in our primary experiments.

Declarations

Competing interests

The authors declare no competing interests.

References

- [1] Hoffmann MA, Nothias L-F, Ludwig M, Fleischauer M, Gentry EC, Witting M, Dorrestein PC, Dührkop K, Böcker S (2022) High-confidence structural annotation of metabolites absent from spectral libraries. *Nat Biotechnol* 40:411–421
- [2] Ruttkies C, Schymanski EL, Wolf S, Hollender J, Neumann S (2016) MetFrag relaunched: incorporating strategies beyond in silico fragmentation. *J Cheminform* 8:3
- [3] Dührkop K, Shen H, Meusel M, Rousu J, Böcker S (2015) Searching molecular structure databases with tandem mass spectra using CSI:FingerID. *Proc Natl Acad Sci USA* 112:12580–12585
- [4] Goldman S, Wohlgend J, Stražar M, Haroush G, Xavier RJ, Coley CW (2023) Annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nat Mach Intell* 5:965–979
- [5] Huber F, van der Burg S, van der Hooft JJJ, Ridder L (2021) MS2DeepScore: a novel deep learning similarity measure to compare tandem mass spectra. *J Cheminform* 13:84
- [6] Bushuiev R, Bushuiev A, Samusevich R, Brungs C, Sivic J, Pluskal T (2025) Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS. *Nat Biotechnol*
- [7] Giné R, Pérez-López I, Badia JM, Capellades J, Yanes O (2026) Benchmarking MS/MS featurization strategies for machine learning-driven metabolite structure annotation. *J Am Soc Mass Spectrom* 37:1550–1561
- [8] de Jonge NF, Chekmeneva E, Schmid R, Joas D, Truong L-J, van der Hooft JJJ, Huber F (2026) Cross ionization mode chemical similarity prediction between tandem mass spectra in metabolomics. *Nat Commun* 17:2483

- [9] Bushuiev R, Bushuiev A, de Jonge NF, Young A, Kretschmer F, Samusevich R, Heirman J, Wang F, Zhang L, Dührkop K, Ludwig M, Haupt NA, Kalia A, Brungs C, Schmid R, Greiner R, Wang B, Wishart DS, Liu L-P, Rousu J, Bittremieux W, Rost H, Mak TD, Hassoun S, Huber F, van der Hooft JJJ, Stravs MA, Böcker S, Sivic J, Pluskal T (2024) MassSpecGym: a benchmark for the discovery and identification of molecules. In: *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*
- [10] Kind T, Fiehn O (2007) Seven Golden Rules for heuristic filtering of molecular formulas obtained by accurate mass spectrometry. *BMC Bioinformatics* 8:105
- [11] Beck AG, Muhoberac M, Randolph CE, Beveridge CH, Wijewardhane PR, Kenttämä HI, Chopra G (2024) Recent developments in machine learning for mass spectrometry. *ACS Meas Sci Au* 4:233–246
- [12] Wang Y, Wei L, Liu L, Xu H, Ling H (2026) How well can off-the-shelf LLMs elucidate molecular structures from mass spectra using chain-of-thought reasoning? *J Cheminform* 18:114
- [13] Tan X (2025) A transformer based generative chemical language AI model for structural elucidation of organic compounds. *J Cheminform* 17:103
- [14] Bui-Thi D, Liu Y, Lippens JL, Laukens K, De Vijlder T (2024) TransExION: a transformer based explainable similarity metric for comparing IONS in tandem mass spectrometry. *J Cheminform* 16:61